Photosynthesis in Higher Plants

1. The scientist who discovered oxygen in 1774 –
1) Ingenhousz
2) Priestly
3) Engelmann
4) Van Neil

2. Which of the following statements is incorrect?
1) Photosynthesis is only a chemical process
2) Photosynthesis is responsible for the release of oxygen into the atmosphere by green plants
3) Green plants carry out photosynthesis
4) None of the above

3. In half leaf experiment, the function of KOH soaked cotton is to –
1) Supply CO_2
2) Absorb O_2
3) Absorb CO_2
4) Take out water from leaf

4. The scientist who showed that only green parts of the plants could release oxygen is –
1) Priestly
2) Ingenhousz
3) Engelmann
4) Van Neil

5. Which of the following algae is used by Engelmann to investigate the action spectrum?
1) Chlamydomonas
2) Spirogyra
3) Cladophora
4) Chlorella

6. S-I: Van Neil inferred the O_2 evolved by green plants comes from H_2O
S-II: Priestley hypothesized that plants restore the air whatever breathing animals and burning candles remove
1) Only S-I is correct
2) Only S-II is correct
3) Both S-I and S-II are correct
4) Both S-I and S-II are incorrect

7. Mark the correct matching pair –

Scientist	Observation
1) Van Neil	Variegated leaf experiment
2) Ingenhousz	Absorption spectra of chlorophyll a
3) Engelmann	Produced first action spectra
4) Priestly	Glucose stored as starch

8. Engelmann placed a green alga in a suspension of aerobic bacteria and illuminated the culture by passing light through a prism. He noticed that bacteria predominantly accumulated in the region of –
1) Violet and green light
2) Blue and red light
3) Orange and yellow light

4) Blue and green light

9. The absorption spectra of which of the following pigments resemble the action spectrum produced by Engelmann?

A. Xanthophyll B. Carotenoids

C. Chlorophyll a D. Chlorophyll b

1) C and A only
2) A and B only
3) Only A
4) C and D only

10. The scientist who discovered that glucose is generally made in green part of plant and stored in form of starch –

1) T.W. Engelmann
2) Von Sachs
3) Van Neil
4) Priestly

11. All are the function of accessory pigments, except –

1) Enable a wider range of wavelength of incoming light
2) Transfer the energy to reaction centre
3) Protect reaction centre from photo-oxidation
4) Release of electron on absorption of light

12. S-I: Light reaction occurs in stroma

S-II: Dark reaction is not directly light driven but is dependent on the products (ATP + NADPH) formed in light reaction

1) Only S-I is correct
2) Only S-II is correct
3) Both S-I and S-II are correct
4) Both S-I and S-II are incorrect

13. Mark the correct matching pair w.r.t. pigments showing colour in chromatogram –

1) Chlorophyll a – Bright yellow green
2) Chlorophyll b – Yellow green
3) Xanthophyll – Green
4) Carotenoids – Bright blue green

14. The light harvesting complex (LHC) consists of –

1) Only chlorophyll b
2) Hundreds of pigment molecules bound to proteins
3) Very few molecules of chlorophyll a
4) Only chlorophyll a and chlorophyll b

15. S-I: Red light can excite electrons in photosystem-I

S-II: PS-I is found only in stroma lamellae

1) Only S-I is correct
2) Only S-II is correct
3) Both S-I and S-II are correct
4) Both S-I and S-II are incorrect

16. How many of the given structures are associated with the thylakoid membrane?

PS II, RuBisCO, LHC, PEPcase, of ATP synthase, Cytochrome b6f, F1 of ATP synthase

1) 2
2) 3
3) 4

4) 5

17. The stroma lamella of chloroplast lacks both-

1) PS I and NADP reductase
2) P_{700} and PS 1
3) NADP reductase and PS II
4) NADP reductase enzyme and P_{700}

18. Among:

P_{680}, NADP reductase, Plastoquinone, Phaeophytin, PS I, Water Splitting Complex, PS II, P_{700}

How many components listed above are part of cyclic ETS?

1) Two
2) Three
3) Five
4) Four

19. The photolysis of water is associated to-

1) PS II
2) PS I
3) Carotenoid
4) P_{700}

20. Which one of the following is formed in non-cyclic photophosphorylation?

1) Only NADP & ATP
2) Only ATP
3) NADH + H^+ and O_2
4) NADPH + H^+, ATP and O_2

21. All are correct statement related to cyclic photophosphorylation, except

1) Only PS I is involved.
2) Synthesis of NADPH + H^+ occurs
3) Only synthesis of ATP occurs.
4) Occurs only when light of wavelength beyond 680 nm is available for excitation.

22. Mark the incorrect matching pair-

1) CFo of ATP synthase - Embedded in the thylakoid membrane
2) Nitrogen - Part of Chl. a structure
3) Phaeophyti n - Reduce NADPH
4) Cyclic photophosp horylation - By product only ATP

23. S-I: PS II absorb at lower energy than PSI

S-II: NADP reductase is not located in granal thylakoid

1) Only S-I is correct
2) Only S-II is correct
3) Both S-I and S-II are correct
4) Both S-I and S-II are incorrect

24. The correct sequence of e^- movement in cyclic photophosphorylation is-

1) PS I - Fes - Ferredoxinplastoquinone- $cytb_6f$- PSI
2) PS I– phaeophytin – plastoquinone – plastocyanin - $cytb_6f$
3) PS I - Fes - plastoquinone- $cytb_6f$ - plastocyanin
4) PS I - Fes - plastoquinone- $cytb_6f$ - plastocyanin- PSI

25. The acceptor of CO_2 during the dark reaction is

1) PGA
2) PEP carboxylase

3) RuBP
4) $NADPH_2$

26. Identify the incorrect matched pair-
1) Regeneration of RuBP in C_3 cycle - Mesophyll cells
2) RuBP oxygenase activity – Chloropla st
3) Photorespiratory loss of CO_2 – Cytoplas m
4) First stable product of C_3 cycle - PGA

27. Which of the following favours Photorespiration?
1) Low light intensity
2) High O_2 and Low CO_2
3) Only low temperature
4) Low O_2 and high CO_2

28. Select the incorrect matched pair-
1) Plants – Maize
2) Calvin cycle – PGA
3) Hatch and Slack cycle – Sorghum
4) C_4 Plant – Karanz Anatomy

29. All are correct for photorespiration, except-
1) Also known as C-2 cycle
2) No synthesis of ATP
3) Primary acceptor of Oxygen is RuBP
4) Primary product is 4C compound

30. The dark reaction of photosynthesis occurs in-
1) Appressed part of grana
2) Stroma lamellae
3) Stroma
4) Outer membrane of thylakoid

31. Kranz anatomy is seen in-
1) Sorghum
2) Maize
3) Sugarcane
4) All of these

32. In which step of the Calvin cycle is both ATP as well as NADPH + H^+ required for the conversion?
1) $RuBP + CO_2 \rightarrow 2 \times PGA$
2) $PGA \rightarrow PGAL$
3) $PGAL \rightarrow DHAP$
4) Fructose 1, 6 biphosphate → Glucose

33. The location of primary carboxylation enzyme in C_4 plants
1) Chloroplast of mesophyll cells
2) Cytoplasm of mesophyll cells
3) Chloroplast of bundle sheath cells
4) Cytoplasm of bundle sheath cells

34. The first stable product of CO_2 fixation in C_4 plant is –
1) PEP
2) PGA

3) PGAL
4) OAA

35. How many times does Calvin cycle needs to be operated for the synthesis of one glucose molecule?
1) 8
2) 4
3) 2
4) 6

36. The dissimilarity between C_4 and C_3 plants is –
1) Type of end products of photosynthesis
2) The primary acceptor of CO_2 during carbon fixation
3) Type of pigments involved in photosynthesis
4) Number of NADPH that are consumed during the starch synthesis process

37. During Hatch and Slack pathway, primary CO_2 fixation occurs in –
1) Large thick-walled cells
2) Mesophyll cells
3) Vascular bundles
4) Bundle sheath cell cytoplasm

38. The correct sequence of steps involved in C_4 cycle is-
1) CO_2 fixation → Transport → Decarboxylation
2) CO_2 → fixation → Decarboxylation → Recarboxylation
3) Transport → Regeneration → CO_2 fixation
4) CO_2 fixation → Regeneration → Transport

39. S-I: C_4 plants take more energy to fix one CO_2.
S-II: Reduction of PGA to form sugar occurs in stroma of chloroplast in mesophyll cells in C_4 plants
1) Only S-I is correct
2) Only S-II is correct
3) Both S-I and S-II are correct
4) Both S-I and S-II are incorrect

40. Mark the incorrect matching pair-
1) Dark reaction in C_4 plant – Mesophyll cells
2) PEPcase in C_4 plant – Mesophyll cells
3) 1 CO_2 fixation in C_3 plant – 3 ATP required
4) 1 CO_2 fixation in C_4 plant – 4 ATP required

41. Number of ATP to form one glucose molecule in C_4 plant is-
1) 60 ATP
2) 30 ATP
3) 36 ATP
4) 45 ATP

42. In day time, which of the following process can occur in C_4 plant?
1) Calvin cycle
2) Photorespiration
3) Cyclic ATP synthesis
4) Both (a) and (c)

43. S-I: Both the mesophyll cell of C_3 plant and bundle sheath cell of C_4 plant have cyclic ATP synthesis.

S-II: In reduction step of Calvin cycle, 2 molecules of ATP are utilized per CO_2 fixation.

1) Only S-I is correct
2) Only S-II is correct
3) Both S-I and S-II are correct
4) Both S-I and S-II are incorrect

44. Which of the following is not incorrect regarding bundle sheath cells of C_4 plant?

1) Bundle sheath present in single layers
2) Intercellular spaces are present between bundle sheath cells
3) Bundle sheath cell thin-walled cell
4) Bundle sheath have impervious wall

45. How many components listed below are not associated with the Calvin cycle?
Oxaloacetic acid, Malic acid, PEP, 3PGA, PEPcase, PSI, PSII, RuBisCO, NADPH

1) Five
2) Four
3) Six
4) Three

Answer Key

1	2	**11**	4	**21**	2	**31**	4	**41**	2
2	1	**12**	2	**22**	3	**32**	2	**42**	4
3	3	**13**	2	**23**	4	**33**	1	**43**	3
4	2	**14**	2	**24**	4	**34**	4	**44**	4
5	3	**15**	1	**25**	3	**35**	4	**45**	3
6	3	**16**	4	**26**	3	**36**	2		
7	3	**17**	3	**27**	2	**37**	2		
8	2	**18**	2	**28**	1	**38**	1		
9	4	**19**	1	**29**	4	**39**	1		
10	2	**20**	4	**30**	3	**40**	4		